

Deep Learning – Pretrained Models

1. Image Classification Models

1. VGG16

Pretrained model developed for image classification tasks.

Link: <https://keras.io/api/applications/vgg/>

2. VGG19

A deeper version of VGG architecture used for image classification.

Link: <https://keras.io/api/applications/vgg/>

3. ResNet50

A deep convolutional neural network commonly used for image classification and feature extraction.

Link: <https://keras.io/api/applications/resnet/>

4. ResNet101

A deeper ResNet architecture used for advanced image classification tasks.

Link: <https://keras.io/api/applications/resnet/>

5. InceptionV3

A pretrained CNN model widely used for image classification and feature extraction.

Link: <https://keras.io/api/applications/inceptionv3/>

6. Xception

A pretrained deep learning model based on depthwise separable convolutions.

Link: <https://keras.io/api/applications/xception/>

7. MobileNetV2

A lightweight pretrained model designed for efficient image classification, especially on mobile and edge devices.

Link: <https://keras.io/api/applications/mobilenet/>

8. EfficientNetB0

An efficient CNN architecture that provides good accuracy with relatively low computational cost.

Link: <https://keras.io/api/applications/efficientnet/>

9. DenseNet121

A pretrained CNN model that uses dense connections between layers for effective feature learning.

Link: <https://keras.io/api/applications/densenet/>

10. ConvNeXt

A modern CNN architecture designed for high-performance image classification.

Link: <https://keras.io/api/applications/convnext/>

2. Object Detection Models

1. YOLO (You Only Look Once)

A popular real-time object detection model used to detect and locate objects in images and videos.

Link: <https://docs.ultralytics.com/models/>

2. YOLOv5

A widely used YOLO version for fast and accurate object detection.

Link: <https://github.com/ultralytics/yolov5>

3. YOLOv8

An advanced YOLO model that supports object detection, image classification, and segmentation.

Link: <https://docs.ultralytics.com/models/yolov8/>

4. YOLOv11

A newer YOLO architecture designed for efficient computer vision tasks including object detection and segmentation.

Link: <https://docs.ultralytics.com/models/yolo11/>

5. Faster R-CNN

A two-stage object detection model that provides accurate object localization and classification.

Link: https://docs.pytorch.org/vision/main/models/faster_rcnn.html

6. DETR (Detection Transformer)

A transformer-based model designed for end-to-end object detection without traditional anchor boxes.

Link: https://huggingface.co/docs/transformers/model_doc/detr

7. RetinaNet

An object detection model designed to handle the problem of detecting objects with different sizes and class imbalance.

Link: <https://docs.pytorch.org/vision/main/models/retinanet.html>

8. SSD (Single Shot MultiBox Detector)

A single-stage object detection model designed for fast object detection.

Link: <https://docs.pytorch.org/vision/main/models/ssd.html>

3. Image Segmentation Models

1. U-Net

A popular deep learning model used for image segmentation, especially in medical image

analysis.

Link: <https://arxiv.org/abs/1505.04597>

2. DeepLabV3

A semantic segmentation model designed to accurately identify and classify pixels in an image.

Link: <https://docs.pytorch.org/vision/main/models/deeplabv3.html>

3. DeepLabV3+

An improved version of DeepLab that provides better object boundary and segmentation accuracy.

Link: <https://github.com/VainF/DeepLabV3Plus-Pytorch>

4. Mask R-CNN

A model that performs both object detection and instance segmentation.

Link: https://docs.pytorch.org/vision/main/models/mask_rcnn.html

5. PSPNet (Pyramid Scene Parsing Network)

A semantic segmentation model that uses pyramid pooling to understand different regions of an image.

Link: <https://github.com/hszhao/PSPNet>

6. SegFormer

A transformer-based model designed for efficient and accurate semantic segmentation.

Link: https://huggingface.co/docs/transformers/model_doc/segformer

7. SAM (Segment Anything Model)

A foundation model capable of performing general-purpose image segmentation using prompts.

Link: <https://github.com/facebookresearch/segment-anything>

4. Natural Language Processing (NLP) Models

1. BERT (Bidirectional Encoder Representations from Transformers)

A pretrained transformer model widely used for text classification, question answering, and language understanding.

Link: https://huggingface.co/docs/transformers/model_doc/bert

2. RoBERTa

An improved version of BERT trained with optimized training techniques for better language understanding.

Link: https://huggingface.co/docs/transformers/model_doc/roberta

3. ALBERT

A lightweight version of BERT designed to reduce memory usage and improve training

efficiency.

Link: https://huggingface.co/docs/transformers/model_doc/albert

4. DistilBERT

A smaller and faster version of BERT that maintains much of BERT's language understanding capability.

Link: https://huggingface.co/docs/transformers/model_doc/distilbert

5. XLNet

A transformer-based pretrained model designed to capture contextual information from text effectively.

Link: https://huggingface.co/docs/transformers/model_doc/xlnet

6. T5 (Text-to-Text Transfer Transformer)

A pretrained model that converts different NLP tasks into a text-to-text format.

Link: https://huggingface.co/docs/transformers/model_doc/t5

7. GPT (Generative Pre-trained Transformer)

A family of pretrained transformer models designed for natural language generation and understanding.

Link: https://huggingface.co/docs/transformers/model_doc/gpt2

8. ELECTRA

A transformer-based pretrained model that uses an efficient pretraining approach for language understanding.

Link: https://huggingface.co/docs/transformers/model_doc/electra

5. Speech and Audio Models

1. Wav2Vec 2.0

A pretrained speech representation model used for automatic speech recognition and other speech-related tasks.

Link: https://huggingface.co/docs/transformers/model_doc/wav2vec2

2. HuBERT

A self-supervised pretrained model for learning useful speech representations from audio data.

Link: https://huggingface.co/docs/transformers/model_doc/hubert

3. Whisper

A pretrained speech recognition model developed for converting speech into text and supporting multilingual speech processing.

Link: <https://github.com/openai/whisper>

4. wavLM

A pretrained speech model designed for speech recognition, speaker identification, and other audio tasks.

Link: https://huggingface.co/docs/transformers/model_doc/wavlm

5. SpeechT5

A pretrained model that supports multiple speech and text processing tasks, including speech synthesis and speech recognition.

Link: https://huggingface.co/docs/transformers/model_doc/speecht5

6. Face Recognition Models

1. FaceNet

A deep learning model that converts face images into numerical representations called embeddings, which can be used for face recognition and verification.

Link: <https://arxiv.org/abs/1503.03832>

2. DeepFace

A deep learning-based face recognition system designed to identify and verify human faces.

Link: https://www.cs.toronto.edu/~ranzato/publications/taigman_cvpr14.pdf

3. ArcFace

A face recognition model that produces highly discriminative face embeddings for accurate face identification and verification.

Link: <https://github.com/deepinsight/insightface>

4. VGGFace

A pretrained deep convolutional neural network designed for face recognition and face representation.

Link: https://www.robots.ox.ac.uk/~vgg/software/vgg_face/

5. VGGFace2

A large-scale face recognition model and dataset designed to improve recognition across different poses and ages.

Link: https://www.robots.ox.ac.uk/~vgg/data/vgg_face2/

7. Image Generation Models

1. GAN (Generative Adversarial Network)

A deep learning architecture used to generate new images that resemble the training data.

Link: <https://arxiv.org/abs/1406.2661>

2. StyleGAN

A GAN-based model designed to generate high-quality and realistic images with control over different image features.

Link: <https://github.com/NVLabs/stylegan>

3. StyleGAN2

An improved version of StyleGAN that produces more realistic and higher-quality images.

Link: <https://github.com/NVLabs/stylegan2>

4. Stable Diffusion

A diffusion-based pretrained model used for generating images from text prompts and other inputs.

Link: https://huggingface.co/docs/diffusers/api/pipelines/stable_diffusion/overview

5. VQGAN

A generative model that combines vector quantization with GANs for high-quality image generation.

Link: <https://github.com/CompVis/taming-transformers>

6. BigGAN

A large-scale GAN architecture designed to generate high-quality and diverse images.

Link: <https://github.com/ajbrock/BigGAN-PyTorch>

8. Video Processing Models

1. VideoMAE

A self-supervised pretrained video model used for video understanding and action recognition tasks.

Link: https://huggingface.co/docs/transformers/model_doc/videomae

2. TimeSformer

A transformer-based pretrained model designed for video classification and understanding.

Link: https://huggingface.co/docs/transformers/model_doc/timesformer

3. Video Swin Transformer

A transformer-based model designed to capture spatial and temporal information from videos.

Link: <https://github.com/SwinTransformer/Video-Swin-Transformer>

4. X3D

A family of efficient video models designed for video classification and action recognition.

Link: <https://pytorchvideo.readthedocs.io/en/stable/api/models/x3d.html>

5. SlowFast Networks

A video recognition architecture that uses different temporal pathways to capture slow and

fast motion information.

Link: <https://github.com/facebookresearch/SlowFast>

9. Multimodal Models

1. CLIP (Contrastive Language–Image Pre-training)

A pretrained model that learns the relationship between images and text. It can be used for image classification, image-text matching, and image search.

Link: <https://github.com/openai/CLIP>

2. BLIP (Bootstrapping Language-Image Pre-training)

A pretrained vision-language model used for image captioning, visual question answering, and image-text understanding.

Link: <https://github.com/salesforce/BLIP>

3. BLIP-2

An improved vision-language model that connects pretrained image and language models for multimodal tasks.

Link: https://huggingface.co/docs/transformers/model_doc/blip-2

4. LLaVA (Large Language and Vision Assistant)

A multimodal model that combines vision and language capabilities to understand and answer questions about images.

Link: <https://github.com/haotian-liu/LLaVA>

5. Flamingo

A multimodal model designed to process and understand both images and text.

Link: <https://deepmind.google/discover/blog/tackling-multiple-tasks-with-a-single-visual-language-model/>